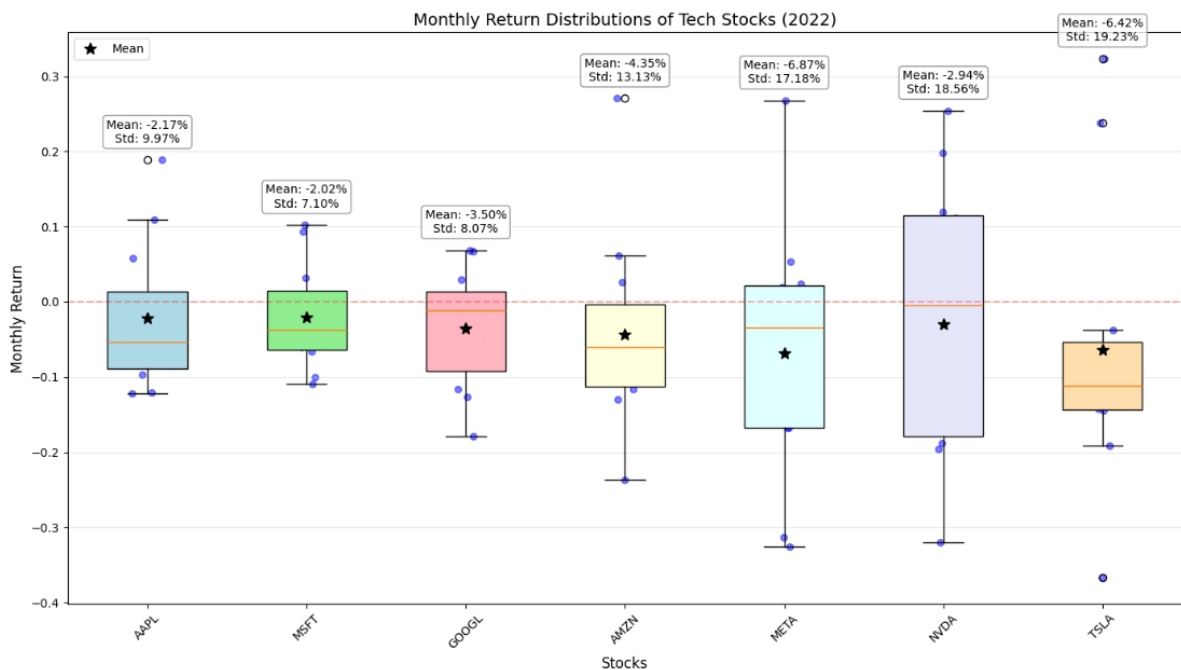




This visualization shows for each stock:

- The median return (middle line in the box)
- The interquartile range (the box itself)
- The range of typical values (the "whiskers")
- Outlier months (individual points beyond the whiskers)

Box plots are excellent for comparing volatility and return characteristics across multiple securities at a glance.

Correlation Analysis: Understanding Market Relationships

Correlations measure how assets move in relation to each other—critical information for portfolio construction, risk management, and pairs trading. Let's examine correlations among major tech 1. stocks:

2. **# Calculate correlation matrix**
3. `correlation_matrix = tech_returns.corr()`
- 4.
5. **# Create a heatmap visualization**